

Luna — Software Requirements Specification

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1. Introduction

Luna connects surplus food with people who need it, before the food goes bad. It links food donors, volunteers and NGOs through one app, and uses AI to check food safety and decide where each donation should go.

The problem. India wastes about ₹92,000 crore of food every year while many people go hungry. Food is not short; coordination is. Restaurants don't know which NGO needs food, NGOs can't see what's available, and no one organises the last-mile pickup.

What Luna does differently. Luna sees both sides of the city at once: where surplus is created and where hunger is. That lets it spot gaps, like an NGO in Jayanagar that gets nothing every Wednesday, and actively fix them.

Scope of this version. A pilot in one city area (Bengaluru), covering restaurants, caterers, households, NGOs, shelters and community fridges. It works on ordinary smartphones and WhatsApp, with no cold-chain needed.

2. Who uses Luna

Luna has four kinds of users, each with their own screens.

User	Who they are	What they want from Luna
Donor	Restaurants, caterers, hotels, event organisers, households	List leftover food in seconds and know it reached someone
Recipient (NGO)	NGOs, shelters, orphanages, old-age homes, community fridges	Get the right food, on time, without chasing donors
Volunteer	Students, delivery riders, local residents	Simple pickup and drop tasks near them, with recognition
Admin	The Luna team, city partners	See the whole city, watch the AI's decisions, step in when needed

3. Feature 1: Food Heat Map

Summary. A live city map with two layers: where surplus food is created, and where hunger is not being met. Where the two don't line up, the map highlights a "gap" that Luna then works to close.

What it shows

- **Surplus layer:** which areas, which days and times, and which kinds of donors (restaurants, caterers, households) give away the most food.
- **Need layer:** where NGOs are, how much food they need, and how much they actually received. Areas with no NGO coverage are shown too.
- **Gap spots:** areas where need is high but little food arrives, shown in red.
- **Time slider:** move through days and hours to see patterns, such as "Wednesday nights are always short in Jayanagar".
- **Area details:** tap any area to see its top donors, its NGOs, and how things changed over the last month.
- **Weekly list:** the top 10 most underserved areas this week.

Where the first data comes from. Before Luna has much activity, the need layer is filled using public information such as registered NGO locations and population density, so the map is useful from day one.

Examples

- *Example 1:* The map shows Koramangala has plenty of restaurant surplus every evening, while Jayanagar, 5 km away, turns red every Wednesday. Luna flags this gap to the Matching Agent automatically.
- *Example 2:* An admin slides the time bar to Sunday afternoon and sees wedding caterers in Rajajinagar produce large surpluses. The team decides to recruit more volunteers there for weekends.
- *Example 3:* A CSR partner looks at the map and chooses to sponsor a community fridge in an area that has no NGO at all.

4. Feature 2: AI Food Checker

Summary. Before any food is sent anywhere, Luna checks whether it is safe to eat and for how long. The donor uploads a photo and a few details; the AI gives back a clear safety verdict.

What the donor provides: 1–3 photos, the type of food, when it was cooked, how it has been stored (room temperature or fridge), and roughly how much there is.

What the checker gives back

- What the food is (rice, curry, bread, fruit, packaged items, and so on)
- A freshness score out of 100
- Any warning signs spotted, such as mould, discolouration or sliminess
- A **"safe until"** time
- Whether it is veg, non-veg or contains egg
- An estimate of how many people it can feed
- A one-line reason in plain words, for example "Cooked 1 hour ago, looks fresh, safe for 6 more hours"
- How sure the AI is about its answer

How it decides. Simple food-safety rules come first (for example, cooked rice kept at room temperature for over 4 hours is always rejected). The photo check then adjusts the verdict. The rules always win if they disagree with the AI.

Quality grades. Every listing gets a grade that decides where it can go.

Grade	Meaning	Who can receive it
A – Premium	Very fresh, safe for 6+ hours	Anyone, including children, elderly and patients
B – Good	Fresh, safe for 3–6 hours	General NGOs and shelters
C – Serve now	Safe for under 3 hours	Only places that can serve it within an hour
D – Not for people	Unsafe or failed a safety rule	Animal shelters or composting only
Unsure	AI isn't confident	Sent, but the volunteer checks it at pickup

Examples

- *Example 1:* A hotel uploads a photo of veg biryani cooked 30 minutes ago. Result: score 88, safe for 7 hours, Grade A, feeds about 50.
- *Example 2:* A caterer lists dal-rice cooked 3 hours ago and left out. Result: score 65, safe for 1.5 more hours, Grade C, "serve immediately".
- *Example 3:* A household uploads bread with visible mould. Result: Grade D, "Not safe for people; mould detected."
- *Example 4:* A blurry photo of paneer curry. Result: score 70, but low confidence, so the volunteer gets a quick smell-and-look checklist at pickup.

5. Feature 3: Smart Matching Agent

Summary. An AI agent that decides where each donation should go and makes it happen. It picks the right NGO, finds a volunteer, plans the route, sends WhatsApp messages, and changes plans if something goes wrong. It works in two ways:

- **When food is listed:** it finds the best recipient and arranges the pickup.
- **When the heat map shows a gap:** it contacts donors near that area ahead of time and asks for food.

Step 1 – Rule out places that can’t safely take the food. A recipient is skipped if any of these fail:

- **Time:** the food must stay safe until the recipient actually serves it, not just until it arrives.
- **Vulnerable groups:** places serving children, elderly, pregnant women or patients only get Grade A food.
- **Storage:** food that needs a fridge only goes to places that have one.
- **Dietary fit:** veg, Jain, halal, no-egg and allergy rules from the NGO’s profile are always respected.
- **Grade D food** never goes to people.

Step 2 – Rank the remaining recipients. The agent weighs how urgent the food is, how close the recipient is, how well the food suits them, how underserved they have been, and whether they have space for that amount. Top-quality food is steered towards the most vulnerable groups.

Step 3 – Make it happen. It offers the food to the top recipient with a countdown. If they don’t respond, it moves to the next. Once accepted, it assigns a volunteer and a route. Shorter-life food gets shorter countdowns and the nearest volunteer.

Step 4 – Keep watching. If a volunteer is delayed and the food would expire, the agent redirects to a closer recipient on the way and tells everyone involved.

Every decision comes with a plain-language reason, visible to the admin.

Examples

- *Fresh food, vulnerable group:* 50 portions of fresh veg biryani (Grade A). The agent sends it to a children’s shelter 4 km away instead of a general NGO 1.5 km away, because the shelter can only take top-quality food.
- *Short time window:* 120 portions of dal-rice safe for 1.5 hours. A Jayanagar NGO that serves dinner at 8 pm is ruled out because the food would expire first. The agent splits it: 80 to a Koramangala shelter that serves on arrival, 40 to a nearby community fridge.

- *Dietary fit:* A listing has 50 veg and 30 chicken portions. The veg portions go to a veg-only NGO; the chicken goes to a different shelter.
- *Traffic delay:* A volunteer gets stuck and the food will expire before reaching the original NGO. The agent redirects to a shelter 2 km away and gives the first NGO priority on the next listing.
- *Gap closing:* The heat map flags Jayanagar as empty every Wednesday. On Wednesday afternoon the agent messages nearby restaurants that often have surplus, and books a volunteer in advance.

6. Feature 4: Fair Share (no NGO left out)

Summary. Luna makes sure no organisation keeps missing out on food just because its requirements are hard to meet. It tracks how long each NGO has gone without food and steps up its efforts the longer they wait. Safety and dietary rules are never bent; only preferences are.

What can and can't be relaxed

Never relaxed	Can be relaxed when an NGO is struggling
Food must be safe until it is served	How far the NGO usually accepts food from
Grade D food never goes to people	Minimum amount the NGO wants per delivery
Vulnerable groups only get Grade A food	Preferred delivery times
Fridge-only food goes only where there's a fridge	Cuisine preferences, such as "rice meals only"
Dietary rules: veg, Jain, halal, no-egg, allergies	Using only the most reliable volunteers

How it works. Each NGO gets a "hunger level" based on how much food it received compared to what it needs, and how long since its last delivery. As the hunger level rises, Luna does more:

1. **Normal:** regular matching.
2. **Boost:** the NGO gets a higher priority in close decisions.
3. **Relax:** preferences are loosened, for example accepting food from farther away or smaller amounts.
4. **Reserve:** new nearby listings are offered to this NGO first, and no single NGO can take most of an area's food.

5. **Hunt:** the agent actively looks for donors who can give exactly what this NGO needs.
6. **Escalate:** if nothing works in 48 hours, a human admin is alerted with the reason.

Learning why an NGO is missing out. Every time an NGO is skipped, Luna notes why (too far, dietary rule, timing, amount). It then picks a fix that targets that reason, like splitting a mixed listing, combining small listings, or finding a volunteer who travels that way. If an NGO's own preferences are blocking it, Luna suggests relaxing them, but never changes dietary rules without the NGO's permission.

Examples

- *Jain ashram:* A Jain ashram has declined 18 listings in 5 days because of onion and garlic. Luna keeps the rule, contacts Jain kitchens, temple kitchens and sweet shops nearby, and routes fruit and suitable packaged food there. Deliveries restart within 2 days.
- *Too far away:* An NGO in Jayanagar only accepts food within 3 km, but most surplus is 5.5 km away in Koramangala. Luna widens the distance for this NGO and finds a volunteer who commutes that way every evening.
- *One NGO taking everything:* A large, fast NGO in HSR Layout is grabbing 70% of the area's listings. A small shelter nearby gets little. Luna limits the big NGO's share and gives the shelter the first 10 minutes on new listings until it catches up.
- *Safety still comes first:* A children's shelter is very hungry, and Grade B food is available 800 m away. Luna still doesn't send it. Instead, it prioritises the shelter for the next Grade A listing and asks nearby donors of freshly cooked food.

7. Feature 5: Apps for Donors, NGOs and Volunteers

Summary. Each user gets simple screens built around the one thing they need to do. Listing food should take under 30 seconds, and every key action should take one tap.

Donor app

- Quick listing: take a photo, the AI fills in most details, confirm and done
- Option to list by just sending a WhatsApp photo and message
- Live status: "Checked → Matched → Picked up → Delivered"
- Impact receipt after each donation: meals served, where it went

Example: A restaurant manager snaps a photo of leftover pulao at 10 pm. The app fills in "veg pulao, about 30 portions". She taps confirm and later gets "Delivered to Hope Shelter, fed 30 people."

NGO app

- Incoming food offers with a countdown timer; accept or decline in one tap

- Post a need, like "60 veg lunches for kids tomorrow"
- A profile with capacity, serving times, fridge availability and dietary rules
- Delivery history and feedback on food quality

Example: An NGO sees "40 portions veg rice, safe until 9 pm, arriving 7:40 pm" with 5 minutes to accept. They tap accept and get the volunteer's name and live location.

Volunteer app

- Nearby task cards showing pickup, drop and time needed
- Turn-by-turn route
- Pickup checklist (and a quick food check when the AI was unsure)
- Confirm handover with a code or QR scan and a photo
- Points, streaks and badges

Example: A student sees a task: "Pickup 1.2 km, drop 2 km, 25 min." She accepts, follows the route, scans the NGO's QR code on delivery and earns 20 points.

Admin dashboard

- The heat map with gap spots
- A live feed of every decision the agent makes, with its reason
- Flagged listings, starving NGOs and escalations needing a human

Example: The admin sees the agent has escalated a shelter that received nothing for 48 hours, and calls a CSR partner to sponsor meals.

8. Additional Features

These features make Luna more trustworthy and keep people coming back. The first two are part of the main build; the rest are planned add-ons.

Feature	What it does	Example
Handover verification (main build)	Each handover is confirmed with a code or QR scan plus a photo, so there's a clear record from donor to NGO	The NGO scans the volunteer's QR; the donor instantly sees "Delivered at 8:05 pm"
Quality feedback (main build)	NGOs rate the food after delivery, and the AI checker learns from these ratings	Several NGOs rate one caterer's "Grade A" food as stale, so the checker becomes stricter for that food type

Feature	What it does	Example
Reliability scores	Donors and volunteers build a track record; reliable volunteers get urgent tasks first	A volunteer with 30 on-time deliveries gets a short-window pickup ahead of a newcomer
Kannada and Hindi, with voice	Donors can list food by speaking in their own language	A dhaba owner says "20 chapati, 10 plate dal" in Kannada, and the listing is filled in
Big event mode	Large surpluses from weddings and events are split across several NGOs and volunteers at once	400 portions after a wedding go to 5 NGOs using 6 volunteers
CSR impact reports	Companies get a report of meals saved and waste avoided through their donations	A tech park cafeteria downloads a monthly report for its sustainability goals
Gamification	Points, streaks, badges and area leaderboards for volunteers	"Top volunteer in Indiranagar this month" badge

9. Donor Incentives

Summary. Restaurants get no tax deduction for donating food: Section 80G only covers money, not food or other goods. So Luna has to create the reasons to donate. The main return for donors is goodwill, and goodwill only counts if it is visible and verifiable. Luna makes it both.

What donors get

Incentive	What it gives the donor	Best for	Example
Legal compliance, done for them	Luna auto-generates the donation label FSSAI requires (food name, source, preparation date, last date to eat, veg or non-veg) and keeps a donation log	All donors	A restaurant never has to write labels by hand, and has a clean record if a question ever comes up

Incentive	What it gives the donor	Best for	Example
Impact certificates	A monthly certificate showing meals donated and waste avoided	Small and mid-size restaurants	"Cafe Mocha donated 620 meals in October" certificate, framed at the counter
"Luna Partner" badge	A badge for the shop window and for Zomato/Swiggy listings	Small restaurants	Customers see the badge and choose that restaurant over the one next door
Public leaderboard	Monthly ranking of the top zero-waste donors in each area	Restaurants that want local visibility	"Top 5 zero-waste restaurants in Koramangala" shared on social media
CSR and ESG impact reports	Verified numbers on meals, kg of food saved and carbon avoided, ready for company reports	Hotel chains, caterers, corporate cafeterias	A tech park cafeteria uses Luna's yearly report in its CSR filing on hunger relief
Surprise bags	Donors sell part of their surplus at a discount through Luna	Bakeries, cafes, sweet shops	A bakery sells 10 end-of-day bags at ₹99 each instead of throwing the bread away

Pitch line. Donating doesn't give restaurants a tax break, so goodwill is their only return, and goodwill only matters if people can see it and trust it. Luna turns every donation into visible, verified goodwill.

Note: the CSR angle for corporates should be confirmed with a chartered accountant before it is presented as a guaranteed benefit.

10. General Requirements

These describe how Luna should behave overall, beyond individual features.

- **Fast:** the food check result appears within a few seconds, and a match is found within a minute of listing.
- **Easy to use:** works on basic smartphones and over WhatsApp; listing takes under 30 seconds.

- **Safe:** safety and dietary rules can never be overridden by the AI; unsafe food never reaches people.
- **Transparent:** every AI decision is recorded with a plain-language reason that admins can review.
- **Human in control:** a new donor's first listing is checked by a person, and anything the AI can't solve is escalated to an admin.
- **Respectful messaging:** donors are not spammed; the agent limits how often it contacts each one.
- **Private:** personal details like phone numbers and addresses are only shared with the people involved in a delivery.
- **Reliable:** if the internet drops, pending tasks are saved and resume when the connection returns.
- **Grows city by city:** built to start in one area and expand without redesign.

11. Future Scope

- **Surplus forecasting:** predict which donors will have surplus and when, so pickups are booked before food is even listed. *Example:* "This hotel usually has 15 kg extra on Friday nights."
- **Surprise bags:** people buy surplus food at a discount, giving donors a small return and another route for good food.
- **Community fridge live status:** show what's inside each community fridge in real time.
- **Partnerships with local government:** share heat map insights with city bodies to plan food programmes.